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Replication and Critical Analysis: Elections, Protest, and Political Trust in Russia

[Code ▼](#)

Moscow Population Survey, 2011 — Full Replication with Extended Diagnostics

Replication Analysis

2026-04-07

1 Abstract

This report provides a full replication and critical analysis of a political science study examining how the December 4, 2011 Russian parliamentary election and the December 10, 2011 Bolotnaya Square protest affected political trust among Moscow residents. Using a before/after natural-experiment design with a cross-sectional survey, the original authors find significant post-protest declines in trust. We replicate the three main tables (descriptive t-tests, OLS main effects, partisan heterogeneity models) and extend the analysis with balance checks, McCrary density tests,

power analysis, anticipation-effect tests, multiple-testing corrections, exogeneity tests, group-composition checks, and a detailed methodological critique. Our replication largely confirms the original findings, though we surface several threats to the causal identification strategy that warrant caution when interpreting the results as causal.

2 Introduction

The December 2011 Russian federal election and the ensuing Bolotnaya Square protest represent a pivotal moment in post-Soviet Russian political history. The Moscow Population Survey, collected from late November through late December 2011, straddles both events, creating a quasi-experimental opportunity to assess whether and how exposure to these events shaped political trust in state institutions. The original study uses a difference-in-means (before/after) design, treating the survey date as a running variable and the two events as cutoffs.

This replication proceeds as follows. Section 2 constructs all variables from the raw Stata data file. Section 3 provides descriptive statistics. Sections 4–6 replicate the three paper tables. Sections 7–12 conduct extended diagnostics. Section 13 provides a systematic methodological critique, and Section 14 documents code-integrity checks.

3 Data Setup and Variable Construction

		Show
## Dataset dimensions: 1550 rows x 292 columns		
		Show
## DATA (day) range: 1 30		
		Show
## MONTH values: 11 12		
		Show
## Period counts:		
		Show
## period1 (pre-election, excl. Dec 25): 291		
		Show
## period1min (excl. election day) : 247		
		Show
## period2 (Dec 5–10) : 417		
		Show

##period2min (excl. protest day): 320

Show

##period3 (Dec 11–25): 842

Show

##electday: 44

Show

##protestday: 97

Show

##dec25: 13

Show

Trust variable ranges after recoding (should be 1–5):

Show

Trust variable ranges

	min	max
trustarmy	1	5
trustpolice	1	5
trustfsb	1	5
trustcourt	1	5
trustmungov	1	5
trustfedgov	1	5
trustduma	1	5
trustprocuracy	1	5
trustun	1	5
trustgovind	1	5
		Show

okrug levels: 1 2 3 4 5 6 7 8 9 10

okrug N per level:

1 2 3 4 5 6 7 8 9 10
107 146 173 200 173 232 202 168 120 29

Show

Partisanship distributions:

Show

ur (0/1/2): 1332 218

Show

apolitical (0-4): 805 745

Show

nopart: 1016 534

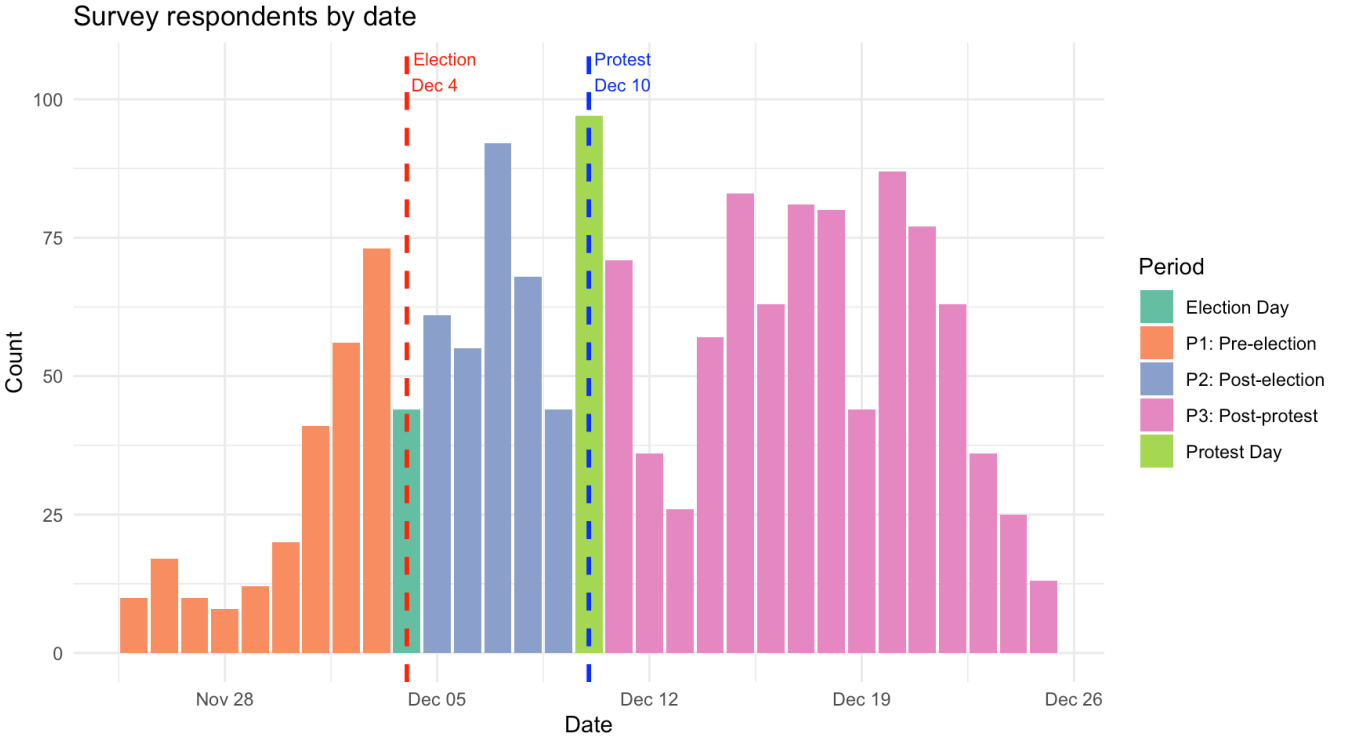
4 Descriptive Statistics

Show

Sample size by period

Period	N
Pre-election (P1: late Nov–Dec 3, excl. Dec 25)	247
Election day (Dec 4)	44
Post-election (P2: Dec 5–9, excl. Dec 10)	320
Protest day (Dec 10)	97
Post-protest (P3: Dec 11–25)	842
Dec 25	13
TOTAL	1550

Show



Show

Summary statistics for trust variables (full sample)

variable	N	Mean	SD	Min	Max
trustarmy	1513	3.130	1.020	1	5
trustpolice	1527	2.780	0.959	1	5
trustfsb	1359	2.968	1.009	1	5
trustcourt	1472	2.711	0.954	1	5
trustmungov	1508	2.816	0.979	1	5
trustfedgov	1510	2.868	1.005	1	5
trustduma	1489	2.588	0.978	1	5
trustprocuracy	1458	2.857	0.973	1	5
trustun	1231	3.010	1.047	1	5
trustgovind	1544	2.842	0.755	1	5

Show

Show

5 Replication of Table 1: Period Comparisons (T-tests)

Table 1 Replication: Period comparisons (t-tests). Positive difference = comparison period higher than reference.

	Variable	Comparison	Mean (ref)	Mean (comp)	Difference	t-stat	p-value	N (ref)	N (comp)	Sig.
P1 vs P2 (pre-election vs. post-election)										
t	trustgovind	P1 vs. P2	2.787	2.671	-0.116	1.752	0.0804	320	247	
t1	trustarmy	P1 vs. P2	3.147	2.898	-0.249	2.796	0.0054	312	245	**
t2	trustpolice	P1 vs. P2	2.771	2.480	-0.291	3.753	0.0002	314	246	***
t3	trustfsb	P1 vs. P2	2.962	2.681	-0.281	3.103	0.0020	288	238	**
t4	trustcourt	P1 vs. P2	2.677	2.550	-0.127	1.525	0.1280	303	242	
t5	trustmungov	P1 vs. P2	2.695	2.680	-0.015	0.178	0.8589	305	244	
t6	trustfedgov	P1 vs. P2	2.720	2.749	0.029	-0.330	0.7415	311	243	
t7	trustduma	P1 vs. P2	2.498	2.529	0.031	-0.363	0.7166	305	242	
t8	trustprocuracy	P1 vs. P2	2.776	2.774	-0.002	0.023	0.9816	303	243	
t9	trustun	P1 vs. P2	2.949	3.073	0.124	-1.318	0.1881	256	219	
P1 vs P3 (pre-election vs. post-protest)										
t10	trustgovind	P1 vs. P3	2.913	2.671	-0.242	4.418	0.0000	837	247	***
t11	trustarmy	P1 vs. P3	3.179	2.898	-0.281	3.687	0.0003	821	245	***
t21	trustpolice	P1 vs. P3	2.862	2.480	-0.382	5.860	0.0000	831	246	***
t31	trustfsb	P1 vs. P3	3.062	2.681	-0.381	5.022	0.0000	714	238	***
t41	trustcourt	P1 vs. P3	2.763	2.550	-0.213	3.087	0.0022	796	242	**
t51	trustmungov	P1 vs. P3	2.907	2.680	-0.226	3.246	0.0013	825	244	**
t61	trustfedgov	P1 vs. P3	2.976	2.749	-0.227	3.080	0.0022	819	243	**
t71	trustduma	P1 vs. P3	2.658	2.529	-0.129	1.759	0.0794	810	242	
t81	trustprocuracy	P1 vs. P3	2.919	2.774	-0.146	2.083	0.0379	780	243	•
t91	trustun	P1 vs. P3	3.042	3.073	0.031	-0.386	0.6995	642	219	
P2 vs P3 (post-election vs. post-protest)										
t12	trustgovind	P2 vs. P3	2.913	2.787	-0.126	2.439	0.0151	837	320	•
t13	trustarmy	P2 vs. P3	3.179	3.147	-0.032	0.466	0.6411	821	312	
t22	trustpolice	P2 vs. P3	2.862	2.771	-0.091	1.430	0.1533	831	314	
t32	trustfsb	P2 vs. P3	3.062	2.962	-0.100	1.397	0.1631	714	288	
t42	trustcourt	P2 vs. P3	2.763	2.677	-0.086	1.292	0.1969	796	303	
t52	trustmungov	P2 vs. P3	2.907	2.695	-0.212	3.231	0.0013	825	305	**
t62	trustfedgov	P2 vs. P3	2.976	2.720	-0.255	3.809	0.0002	819	311	***
t72	trustduma	P2 vs. P3	2.658	2.498	-0.160	2.496	0.0129	810	305	•
t82	trustprocuracy	P2 vs. P3	2.919	2.776	-0.144	2.146	0.0323	780	303	•
t92	trustun	P2 vs. P3	3.042	2.949	-0.093	1.197	0.2321	642	256	

6 Replication of Table 2: OLS Main Results

Show

Show

Table 2 Replication: OLS models with robust SEs (HC1). Key period coefficients only. *** p

Outcome	period3	period2min	protestday	electday
trustgovind	0.158** (0.058)	0.062 (0.067)	0.177* (0.083)	-0.043 (0.117
trustarmy	0.227** (0.079)	0.282** (0.092	0.504*** (0.11	) -0.085 (0.164
trustpolice	0.285*** (0.069)	0.242** (0.082	0.377*** (0.10	) 0.222 (0.156)
trustfsb	0.273*** (0.076)	0.185* (0.09)	0.282* (0.123)	0.211 (0.174)
trustcourt	0.13. (0.073)	0.084 (0.086)	0.201. (0.11)	0.108 (0.154)
trustmungov	0.172* (0.073)	-0.04 (0.084)	0.138 (0.124)	-0.207 (0.145
trustfedgov	0.174* (0.077)	-0.054 (0.089)	0.065 (0.122)	-0.282. (0.15
trustduma	0.052 (0.078)	-0.06 (0.087)	-0.089 (0.123)	-0.208 (0.141
trustprocuracy	-0.013 (0.076)	-0.141 (0.088)	-0.121 (0.121)	-0.044 (0.149
trustun	0.051 (0.085)	-0.106 (0.097)	0.048 (0.126)	-0.486** (0.1

Show

N observations per model:

Show

trustgovind trustarmy trustpolice trustfsb trustcourt 1521 1491 1505 1338 1450 trustmungov trustfedgov trustduma
trustprocuracy trustun 1485 1487 1466 1437 1212

7 Replication of Table 3: Partisan Heterogeneity Models

Show

Show

Table 3 Replication: Partisan heterogeneity models (OLS, robust SEs). *** p

Outcome	period3	period2min	protestday	electday	ur	urafterprotest	urpostelection	apolitical	apolafterprotest	apolpostelection
trustgovind	0.195* (0.08)	0.052 (0.099)	0.183* (0.082)	-0.002 (0.113	0.559*** (0.	3. -0.153 (0.	34. -0.017 (0	162. 0.127 (	.08) -0.011 (0	95. 0.055 (0
trustarmy	0.258* (0.106)	0.268* (0.129	0.496*** (0.11	) -0.053 (0.159	0.383* (0.15	) -0.022 (0.	87. 0.078 (0.	38. 0.169 (	.111) -0.051 (0	133. 0.023 (0

Outcome	period3	period2min	protestday	electday	ur	urafterprotest	urpostelection	apolitical	apolafterprotest	apolpostelection
trustpolice	0.255** (0.097)	0.282* (0.118)	0.384*** (0.10)	) 0.263. (0.149)	0.562*** (0.	4. -0.089 (0.	77. 0.006 (0.	11. 0.139 (	.097) 0.112 (0.	2. - 0. 05 7
trustfsb	0.297** (0.112)	0.251. (0.14)	0.288* (0.122)	0.254 (0.165)	0.647*** (0.	43. -0.187 (0.	89. -0.188 (0	224. 0.143 (	.113) 0.036 (0.	36. - 0. 04 9
trustcourt	0.176. (0.103)	0.099 (0.132)	0.192. (0.111)	0.143 (0.151)	0.415** (0.1	3. -0.184 (0.	82. -0.086 (0	221. 0.17 (0	105. -0.026 (0	126. 0.019 (0
trustmungov	0.241* (0.101)	-0.041 (0.123)	0.145 (0.12)	-0.158 (0.151)	0.688*** (0.	32. -0.187 (0.	71. -0.145 (0	211. 0.088 (	.106) -0.065 (0	127. 0.084 (0
trustfedgov	0.182. (0.109)	-0.113 (0.13)	0.087 (0.119)	-0.234 (0.157)	0.671*** (0.	41. -0.07 (0.1	8. 0.157 (0.	9. 0.074 (	.109) 0.03 (0.1	1. 0. 12 6
trustduma	0.146 (0.108)	-0.112 (0.124)	-0.087 (0.119)	-0.157 (0.146)	0.689*** (0.	58. -0.26 (0.1	3. 0.035 (0.	16. 0.122 (	.107) -0.095 (0	129. 0.143 (0
trustprocuracy	0.059 (0.106)	-0.125 (0.132)	-0.123 (0.123)	0.001 (0.147)	0.517*** (0.	5. -0.251 (0.	9. -0.157 (0	242. 0.186. (	0.108) -0.055 (0	129. 0.037 (0
trustun	0.043 (0.119)	-0.033 (0.136)	0.059 (0.128)	-0.476** (0.1	8. 0.217 (0.162	-0.151 (0.	15. -0.253 (0	257. -0.044 (	0.12) 0.077 (0.	5. - 0. 06 8

8 Balance Checks

- Show
- Show
- Show

Balance checks across periods (t-tests, SMDs). Significant p

	Variable	Comparison	Mean (ref)	Mean (comp)	Diff	t-stat	p-value	SMD	Sig.
P1 vs. P2									
t	age	P1 vs. P2	43.513	42.239	-1.274	0.987	0.3239	-0.083	
t1	male	P1 vs. P2	0.459	0.433	-0.026	0.621	0.5348	-0.053	
t2	education	P1 vs. P2	5.369	5.551	0.182	-1.229	0.2196	0.105	
t3	wealth	P1 vs. P2	3.797	3.728	-0.068	1.167	0.2438	-0.099	
t4	permres	P1 vs. P2	0.959	0.935	-0.024	1.258	0.2089	-0.110	
t5	nationality	P1 vs. P2	0.859	0.907	0.048	-1.768	0.0776	0.146	
t6	unemp	P1 vs. P2	0.053	0.069	0.016	-0.768	0.4430	0.066	
t7	statesector	P1 vs. P2	0.044	0.016	-0.028	1.969	0.0495	-0.157	•
t8	ur	P1 vs. P2	0.138	0.170	0.033	-1.058	0.2904	0.091	
t9	comp	P1 vs. P2	0.116	0.158	0.042	-1.441	0.1504	0.124	
t10	opposition	P1 vs. P2	0.169	0.138	-0.031	1.024	0.3063	-0.086	
t11	ldpr	P1 vs. P2	0.100	0.101	0.001	-0.048	0.9621	0.004	
t12	sr	P1 vs. P2	0.116	0.097	-0.018	0.709	0.4784	-0.059	
t13	yab	P1 vs. P2	0.044	0.024	-0.019	1.290	0.1975	-0.105	
t14	pat	P1 vs. P2	0.006	0.008	0.002	-0.256	0.7982	0.022	
t15	rc	P1 vs. P2	0.003	0.008	0.005	-0.763	0.4456	0.068	
t16	dk	P1 vs. P2	0.131	0.170	0.039	-1.271	0.2042	0.109	
t17	nopart	P1 vs. P2	0.347	0.263	-0.084	2.163	0.0310	-0.181	•
P1 vs. P3									
t18	age	P1 vs. P3	43.557	42.239	-1.318	1.190	0.2348	-0.083	
t19	male	P1 vs. P3	0.482	0.433	-0.049	1.361	0.1742	-0.098	
t21	education	P1 vs. P3	5.107	5.551	0.444	-3.493	0.0005	0.260	***
t31	wealth	P1 vs. P3	3.623	3.728	0.105	-2.103	0.0361	0.148	•
t41	permres	P1 vs. P3	0.956	0.935	-0.021	1.211	0.2269	-0.097	
t51	nationality	P1 vs. P3	0.904	0.907	0.003	-0.146	0.8841	0.010	
t61	unemp	P1 vs. P3	0.078	0.069	-0.010	0.514	0.6078	-0.036	
t71	statesector	P1 vs. P3	0.025	0.016	-0.009	0.904	0.3666	-0.058	
t81	ur	P1 vs. P3	0.135	0.170	0.035	-1.298	0.1952	0.099	
t91	comp	P1 vs. P3	0.122	0.158	0.036	-1.376	0.1697	0.106	
t101	opposition	P1 vs. P3	0.151	0.138	-0.013	0.523	0.6012	-0.037	
t111	ldpr	P1 vs. P3	0.101	0.101	0.000	-0.012	0.9904	0.001	
t121	sr	P1 vs. P3	0.101	0.097	-0.004	0.176	0.8607	-0.013	
t131	yab	P1 vs. P3	0.039	0.024	-0.015	1.254	0.2103	-0.080	
t141	pat	P1 vs. P3	0.006	0.008	0.002	-0.343	0.7320	0.027	
t151	rc	P1 vs. P3	0.005	0.008	0.003	-0.541	0.5889	0.045	
t161	dk	P1 vs. P3	0.129	0.170	0.041	-1.526	0.1279	0.117	

	Variable	Comparison	Mean (ref)	Mean (comp)	Diff	t-stat	p-value	SMD	Sig.
t171	nopart	P1 vs. P3	0.361	0.263	-0.098	3.003	0.0028	-0.207	**
P2 vs. P3									
t20	age	P2 vs. P3	43.557	43.513	-0.045	0.043	0.9653	-0.003	
t110	male	P2 vs. P3	0.482	0.459	-0.023	0.696	0.4870	-0.046	
t22	education	P2 vs. P3	5.107	5.369	0.262	-2.340	0.0196	0.155	•
t32	wealth	P2 vs. P3	3.623	3.797	0.174	-3.703	0.0002	0.242	***
t42	permres	P2 vs. P3	0.956	0.959	0.003	-0.253	0.8004	0.016	
t52	nationality	P2 vs. P3	0.904	0.859	-0.044	2.023	0.0436	-0.143	•
t62	unemp	P2 vs. P3	0.078	0.053	-0.025	1.618	0.1060	-0.098	
t72	statesector	P2 vs. P3	0.025	0.044	0.019	-1.487	0.1378	0.110	
t82	ur	P2 vs. P3	0.135	0.138	0.002	-0.093	0.9257	0.006	
t92	comp	P2 vs. P3	0.122	0.116	-0.007	0.317	0.7517	-0.021	
t102	opposition	P2 vs. P3	0.151	0.169	0.018	-0.736	0.4618	0.049	
t112	ldpr	P2 vs. P3	0.101	0.100	-0.001	0.048	0.9616	-0.003	
t122	sr	P2 vs. P3	0.101	0.116	0.015	-0.709	0.4787	0.048	
t132	yab	P2 vs. P3	0.039	0.044	0.005	-0.344	0.7313	0.023	
t142	pat	P2 vs. P3	0.006	0.006	0.000	-0.061	0.9517	0.004	
t152	rc	P2 vs. P3	0.005	0.003	-0.002	0.414	0.6787	-0.025	
t162	dk	P2 vs. P3	0.129	0.131	0.002	-0.081	0.9354	0.005	
t172	nopart	P2 vs. P3	0.361	0.347	-0.014	0.452	0.6517	-0.030	

Show

Show

Omnibus MANOVA tests (Pillai’s trace) for balance across periods

Comparison	Pillai	F_value	df1	df2	p_value
P1 vs. P2	NA	NA	NA	NA	NA
P1 vs. P3	NA	NA	NA	NA	NA
P2 vs. P3	NA	NA	NA	NA	NA

9 McCrary Density Test

The McCrary (2008) density test checks whether respondents “sort” around the cutoff dates — i.e., whether the density of survey respondents is discontinuous at December 4 (election) or December 10 (protest). Under the as-if-random assignment assumption, we should observe no manipulation around the cutoffs.

Show

```
## Continuous date variable (days since Nov 25):
```

[Show](#)

```
## Nov 25 = 0
```

[Show](#)

```
## Dec 04 = 9 (election)
```

[Show](#)

```
## Dec 10 = 15 (protest)
```

[Show](#)

```
## Dec 25 = 30
```

[Show](#)

```
##  
## Election cutoff (cont_date): 9
```

[Show](#)

```
## Protest cutoff (cont_date): 15
```

[Show](#)

```
## === McCrary / Density Test at Election Cutoff (Dec 4) ===
```

[Show](#)

```
##  
## Manipulation testing using local polynomial density estimation.  
##  
## Number of obs =      1550  
## Model =             unrestricted  
## Kernel =            triangular  
## BW method =         estimated  
## VCE method =        jackknife  
##  
## c = 9                Left of c          Right of c  
## Number of obs       247                1303  
## Eff. Number of obs  247                1303  
## Order est. (p)       2                  2  
## Order bias (q)       3                  3  
## BW est. (h)          21                 21  
##  
## Method               T                  P > |T|  
## Robust                -2.43             0.0151
```

```
##
## P-values of binomial tests (H0: p=0.5).
##
## Window Length / 2      <c      >=c      P>|T|
## 1.000                  73       105       0.0199
## 2.000                  129      160       0.0774
## 3.000                  170      252       0.0001
## 4.000                  190      320       0.0000
## 5.000                  202      364       0.0000
## 6.000                  210      461       0.0000
## 7.000                  220      532       0.0000
## 8.000                  237      568       0.0000
## 9.000                  247      594       0.0000
## 10.000                 247      651       0.0000
## NULL
```

[Show](#)

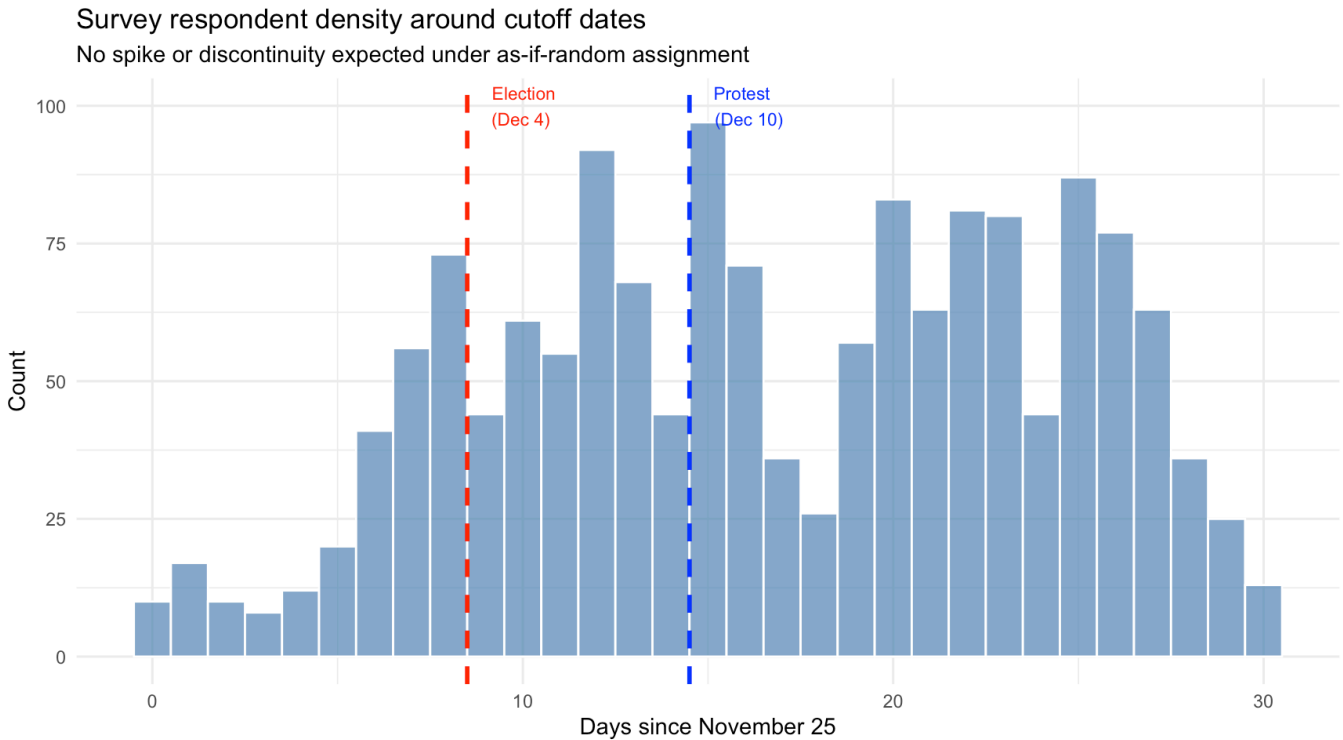
```
## === McCrary / Density Test at Protest Cutoff (Dec 10) ===
```

[Show](#)

```
##
## Manipulation testing using local polynomial density estimation.
##
## Number of obs =      1550
## Model =          unrestricted
## Kernel =         triangular
## BW method =      estimated
## VCE method =     jackknife
##
## c = 15           Left of c           Right of c
## Number of obs    611                 939
## Eff. Number of obs 611                 939
## Order est. (p)    2                   2
## Order bias (q)    3                   3
## BW est. (h)       15                  15
##
## Method           T                   P > |T|
## Robust           -1.8151             0.0695
```

```
##
## P-values of binomial tests (H0: p=0.5).
##
## Window Length / 2      <c      >=c      P>|T|
## 1.000                   44       168      0.0000
## 2.000                   112      204      0.0000
## 3.000                   204      230      0.2301
## 4.000                   259      287      0.2479
## 5.000                   320      370      0.0620
## 6.000                   364      433      0.0160
## 7.000                   437      514      0.0137
## 8.000                   493      594      0.0024
## 9.000                   534      638      0.0026
## 10.000                  554      725      0.0000
## NULL
```

Show



Show

```
## Group sizes:
```

Show

```
## P1 (pre-election): 247
```

Show

P2 (post-election): 320

[Show](#)

P3 (post-protest): 842

[Show](#)

UR supporters (ur>=1): 218

[Show](#)

Non-UR (ur==0): 1332

[Show](#)

sigma(trustgovind): 0.755

[Show](#)##
Observed effects on trustgovind:[Show](#)

period3 (post-protest): 0.158

[Show](#)

period2min (post-election): 0.062

[Show](#)

electday: -0.043

[Show](#)

Minimum Detectable Effects (MDE) and achieved power for trustgovind. MDEs are in trust-scale units.

Comparison	n1	n2	sigma	MDE_alpha05	MDE_alpha005	Power_obs_p3_05	Power_obs_p3_005
P1 vs P2 (full sample)	247	320	0.7554706	0.179	0.233	0.697	0.371
P1 vs P3 (full sample)	247	842	0.7554706	0.153	0.199	0.826	0.537
P2 vs P3 (full sample)	320	842	0.7554706	0.139	0.181	0.891	0.651
P1 vs P3 (UR subgroup)	34	84	0.7554706	0.430	0.560	0.178	0.038

Comparison	n1	n2	sigma	MDE_alpha05	MDE_alpha005	Power_obs_p3_05	Power_obs_p3_005
P1 vs P3 (non-UR subgroup)	208	516	0.7554706	0.174	0.226	0.724	0.400

[Show](#)

MDEs and achieved power per outcome variable (P1 vs P3, trustgovind as reference SD). Power_05/Power_005 = power at alpha=0.05/0.005.

Outcome	sigma	obs_coef_p3	obs_p	MDE_05	MDE_005	Power_05	Power_005
trustgovind	0.755	0.158	0.0060	0.153	0.199	0.824	0.534
trustarmy	1.020	0.227	0.0042	0.207	0.269	0.868	0.606
trustpolice	0.959	0.285	0.0000	0.194	0.253	0.984	0.903
trustfsb	1.009	0.273	0.0003	0.205	0.266	0.962	0.824
trustcourt	0.954	0.130	0.0747	0.193	0.252	0.469	0.178
trustmungov	0.979	0.172	0.0187	0.198	0.258	0.680	0.352
trustfedgov	1.005	0.174	0.0248	0.204	0.265	0.667	0.339
trustduma	0.978	0.052	0.5009	0.198	0.258	0.114	0.019
trustprocuracy	0.973	-0.013	0.8618	0.197	0.257	0.054	0.006
trustun	1.047	0.051	0.5476	0.212	0.276	0.103	0.017

11 Anticipation Effects Test

The December 10 protest was publicly announced on **December 7, 2011**. If respondents interviewed on December 7–9 (Period 2B) already anticipated the protest, their responses may reflect anticipation rather than pure post-election sentiment. We split Period 2 into: - **P2A**: December 5–6 (pre-announcement) - **P2B**: December 7–9 (post-announcement, pre-protest)

[Show](#)

P2A (Dec 5–6): 116 respondents

[Show](#)

P2B (Dec 7–9): 204 respondents

[Show](#)

Orig P2min: 320 respondents

[Show](#)

Anticipation effects test: Comparing P2A (pre-announcement) vs P2B (post-announcement) effects vs. original P2. Reference period = P1 in all cases.

Outcome	Coef	SE	p	sig	Period_label	N
trustarmy	0.282	0.092	0.0022	**	P2 (original: Dec 5-9)	NA
trustarmy	0.328	0.127	0.0099	**	P2A: Dec 5-6 (pre-announcement)	351
trustarmy	0.372	0.123	0.0027	**	P2B: Dec 7-9 (post-announcement)	439
trustcourt	0.084	0.086	0.3289		P2 (original: Dec 5-9)	NA
trustcourt	0.179	0.122	0.1419		P2A: Dec 5-6 (pre-announcement)	346
trustcourt	0.142	0.109	0.1950		P2B: Dec 7-9 (post-announcement)	428
trustduma	-0.060	0.087	0.4921		P2 (original: Dec 5-9)	NA
trustduma	0.162	0.121	0.1814		P2A: Dec 5-6 (pre-announcement)	343
trustduma	-0.135	0.109	0.2155		P2B: Dec 7-9 (post-announcement)	433
trustfedgov	-0.054	0.089	0.5402		P2 (original: Dec 5-9)	NA
trustfedgov	0.209	0.122	0.0868		P2A: Dec 5-6 (pre-announcement)	349
trustfedgov	-0.113	0.115	0.3289		P2B: Dec 7-9 (post-announcement)	435
trustfsb	0.185	0.090	0.0398	•	P2 (original: Dec 5-9)	NA
trustfsb	0.426	0.137	0.0021	**	P2A: Dec 5-6 (pre-announcement)	336
trustfsb	0.163	0.112	0.1451		P2B: Dec 7-9 (post-announcement)	415
trustgovind	0.062	0.067	0.3540		P2 (original: Dec 5-9)	NA
trustgovind	0.217	0.095	0.0224	•	P2A: Dec 5-6 (pre-announcement)	357
trustgovind	0.072	0.088	0.4130		P2B: Dec 7-9 (post-announcement)	444
trustmungov	-0.040	0.084	0.6381		P2 (original: Dec 5-9)	NA
trustmungov	0.218	0.112	0.0529		P2A: Dec 5-6 (pre-announcement)	347
trustmungov	-0.081	0.109	0.4577		P2B: Dec 7-9 (post-announcement)	433
trustpolice	0.242	0.082	0.0034	**	P2 (original: Dec 5-9)	NA
trustpolice	0.440	0.120	0.0003	***	P2A: Dec 5-6 (pre-announcement)	355
trustpolice	0.176	0.101	0.0832		P2B: Dec 7-9 (post-announcement)	439
trustprocuracy	-0.141	0.088	0.1100		P2 (original: Dec 5-9)	NA
trustprocuracy	-0.067	0.125	0.5928		P2A: Dec 5-6 (pre-announcement)	346
trustprocuracy	0.024	0.118	0.8412		P2B: Dec 7-9 (post-announcement)	432
trustun	-0.106	0.097	0.2728		P2 (original: Dec 5-9)	NA
trustun	-0.191	0.138	0.1673		P2A: Dec 5-6 (pre-announcement)	306
trustun	0.209	0.111	0.0611		P2B: Dec 7-9 (post-announcement)	377

12 Multiple Testing Corrections

[Show](#)

Multiple testing corrections for key period coefficients across Tables 2 and 3. Bonferroni and Benjamini-Hochberg corrections applied within each Table x Predictor combination.

Table	Predictor	Outcome	Coef	SE	Raw p	Bonferroni p	BH p	Sig (raw)	Sig (Bonf)	Sig (BH)
Table 2	period2min	trustarmy	0.282	0.092	0.0022	0.022	0.0170000	Yes	Yes	Yes
Table 2	period2min	trustcourt	0.084	0.086	0.3289	1.000	0.5057143	No	No	No
Table 2	period2min	trustduma	-0.060	0.087	0.4921	1.000	0.6002222	No	No	No
Table 2	period2min	trustfedgov	-0.054	0.089	0.5402	1.000	0.6002222	No	No	No
Table 2	period2min	trustfsb	0.185	0.090	0.0398	0.398	0.1326667	Yes	No	No
Table 2	period2min	trustgovind	0.062	0.067	0.3540	1.000	0.5057143	No	No	No
Table 2	period2min	trustmungov	-0.040	0.084	0.6381	1.000	0.6381000	No	No	No
Table 2	period2min	trustpolice	0.242	0.082	0.0034	0.034	0.0170000	Yes	Yes	Yes
Table 2	period2min	trustprocuracy	-0.141	0.088	0.1100	1.000	0.2750000	No	No	No
Table 2	period2min	trustun	-0.106	0.097	0.2728	1.000	0.5057143	No	No	No
Table 2	period3	trustarmy	0.227	0.079	0.0042	0.042	0.0140000	Yes	Yes	Yes
Table 2	period3	trustcourt	0.130	0.073	0.0747	0.747	0.1067143	No	No	No
Table 2	period3	trustduma	0.052	0.078	0.5009	1.000	0.6084444	No	No	No
Table 2	period3	trustfedgov	0.174	0.077	0.0248	0.248	0.0413333	Yes	No	Yes
Table 2	period3	trustfsb	0.273	0.076	0.0003	0.003	0.0015000	Yes	Yes	Yes
Table 2	period3	trustgovind	0.158	0.058	0.0060	0.060	0.0150000	Yes	No	Yes
Table 2	period3	trustmungov	0.172	0.073	0.0187	0.187	0.0374000	Yes	No	Yes
Table 2	period3	trustpolice	0.285	0.069	0.0000	0.000	0.0000000	Yes	Yes	Yes
Table 2	period3	trustprocuracy	-0.013	0.076	0.8618	1.000	0.8618000	No	No	No
Table 2	period3	trustun	0.051	0.085	0.5476	1.000	0.6084444	No	No	No
Table 3	period2min	trustarmy	0.268	0.129	0.0383	0.383	0.1915000	Yes	No	No
Table 3	period2min	trustcourt	0.099	0.132	0.4498	1.000	0.6425714	No	No	No
Table 3	period2min	trustduma	-0.112	0.124	0.3688	1.000	0.6396667	No	No	No
Table 3	period2min	trustfedgov	-0.113	0.130	0.3838	1.000	0.6396667	No	No	No
Table 3	period2min	trustfsb	0.251	0.140	0.0737	0.737	0.2456667	No	No	No
Table 3	period2min	trustgovind	0.052	0.099	0.5992	1.000	0.7490000	No	No	No
Table 3	period2min	trustmungov	-0.041	0.123	0.7373	1.000	0.8084000	No	No	No
Table 3	period2min	trustpolice	0.282	0.118	0.0170	0.170	0.1700000	Yes	No	No
Table 3	period2min	trustprocuracy	-0.125	0.132	0.3429	1.000	0.6396667	No	No	No
Table 3	period2min	trustun	-0.033	0.136	0.8084	1.000	0.8084000	No	No	No
Table 3	period3	trustarmy	0.258	0.106	0.0152	0.152	0.0346000	Yes	No	Yes
Table 3	period3	trustcourt	0.176	0.103	0.0883	0.883	0.1380000	No	No	No
Table 3	period3	trustduma	0.146	0.108	0.1753	1.000	0.2191250	No	No	No
Table 3	period3	trustfedgov	0.182	0.109	0.0966	0.966	0.1380000	No	No	No
Table 3	period3	trustfsb	0.297	0.112	0.0082	0.082	0.0346000	Yes	No	Yes
Table 3	period3	trustgovind	0.195	0.080	0.0147	0.147	0.0346000	Yes	No	Yes
Table 3	period3	trustmungov	0.241	0.101	0.0173	0.173	0.0346000	Yes	No	Yes
Table 3	period3	trustpolice	0.255	0.097	0.0086	0.086	0.0346000	Yes	No	Yes

Table	Predictor	Outcome	Coef	SE	Raw p	Bonferroni p	BH p	Sig (raw)	Sig (Bonf)	Sig (BH)
Table 3	period3	trustprocuracy	0.059	0.106	0.5803	1.000	0.6447778	No	No	No
Table 3	period3	trustun	0.043	0.119	0.7171	1.000	0.7171000	No	No	No

13 Exogeneity Tests

The as-if-random assignment assumption requires that observable characteristics of respondents are not systematically related to the interview date within periods.

Show

Exogeneity tests: OLS of survey date on covariates within each period. Low R2 and high p-value support as-if-random assignment within periods.

Period	N	R2	F_stat	p_value
P1 (pre-election)	243	0.0800	6.471	0.0001
P2 (post-election)	315	0.0204	1.571	0.1819
P3 (post-protest)	828	0.0019	0.387	0.8183

Show

```
##
## --- Within-period exogeneity: P1 (pre-election) (N = 243 ) ---
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -12.5377    4.4600 -2.8112  0.0053 **
## age          0.0858    0.0495  1.7321  0.0846 .
## male        -0.1052    1.5030 -0.0700  0.9443
## education    0.6955    0.4738  1.4681  0.1434
## wealth       4.0681    1.0395  3.9136  0.0001 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## --- Within-period exogeneity: P2 (post-election) (N = 315 ) ---
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)   5.6663    0.5550 10.2100 <2e-16 ***
## age           0.0055    0.0050  1.1024  0.2711
## male          0.1169    0.1464  0.7991  0.4249
## education     0.0041    0.0429  0.0962  0.9234
## wealth        0.2530    0.1129  2.2409  0.0257 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## --- Within-period exogeneity: P3 (post-protest) (N = 828 ) ---
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  18.1864    0.9467 19.2098 <2e-16 ***
## age          -0.0040    0.0090 -0.4420  0.6586
## male         0.1485    0.2622  0.5664  0.5713
## education    0.0384    0.0825  0.4653  0.6418
## wealth      -0.2194    0.1968 -1.1147  0.2653
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

14 Group Composition Issues

[Show](#)

Share of United Russia supporters (ur >= 1) by period

period_label	N	ur_mean	ur_sd
--------------	---	---------	-------

period_label	N	ur_mean	ur_sd
P1	247	0.170	0.376
P2	320	0.138	0.345
P3	842	0.135	0.342

[Show](#)

```
##
## Chi-square test for UR composition across periods:
```

[Show](#)

```
##
## Pearson's Chi-squared test
##
## data: period_ur_mat
## X-squared = 1.9496, df = 2, p-value = 0.3773
```

[Show](#)

```
##
## === Interaction: wealth ~ period_label * ur_any ===
```

[Show](#)

```
##
## t test of coefficients:
##
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.7214    0.0458  81.2444  <2e-16 ***
## factor(period_label)P2      0.0904    0.0624   1.4487   0.1476
## factor(period_label)P3     -0.0817    0.0532  -1.5366   0.1246
## factor(ur_any)1           0.0405    0.1292   0.3136   0.7538
## factor(period_label)P2:factor(ur_any)1 -0.1478    0.1746  -0.8462   0.3976
## factor(period_label)P3:factor(ur_any)1 -0.1623    0.1485  -1.0930   0.2746
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

[Show](#)

```
##
## === Interaction: education ~ period_label * ur_any ===
```

[Show](#)

```
##
## t test of coefficients:
##
##
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      5.6488    0.1230 45.9387  <2e-16 ***
## factor(period_label)P2    -0.2502    0.1619 -1.5453   0.1225
## factor(period_label)P3    -0.5252    0.1375 -3.8195   0.0001 ***
## factor(ur_any)1          -0.5774    0.2993 -1.9290   0.0539 .
## factor(period_label)P2:factor(ur_any)1    0.3606    0.3829   0.9418   0.3465
## factor(period_label)P3:factor(ur_any)1    0.4537    0.3523   1.2879   0.1980
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Methodological Critique

14.1 Estimand and Causal Identification

The paper frames its findings as evidence that elections and protests causally affect political trust. The design is a cross-sectional survey spanning a period that straddles two events, with respondent assignment to interview date treated as quasi-random within periods. The estimand is therefore an **average treatment effect (ATE)** of “being surveyed after the election/protest” on expressed trust, conditional on covariates and district fixed effects.

For this to be causal, the critical assumption is that the date of interview is **as-if randomly assigned** — that is, individual characteristics do not systematically predict which day within or across periods a respondent was surveyed. This is a strong assumption. Survey fieldwork involves quota sampling by district, meaning enumerators work through a district over a span of days; early completers within a district may differ systematically from later completers. Our exogeneity tests (Section 11) probe this assumption directly.

A secondary identification concern is the **stable unit treatment value assumption (SUTVA)**: if news of the election results or the protest announcement spread through social networks before the interview date, respondents technically assigned to one period may have been “treated” by information from the next period. This is particularly acute for Period 2 (December 5–9), where respondents interviewed on or after December 7 could have been aware of the protest announcement.

14.2 Survey Timing and Contamination of Period 2

The December 10 Bolotnaya Square protest was publicly called on December 7 via social media and was widely covered in Russian media by that evening. This means the “post-election only” window effectively consists of only December 5–6 (two days), not the five days (December 5–9) treated as a homogeneous post-election period. Our anticipation effects analysis (Section 9) directly tests whether the Period 2 coefficient is driven by the December 7–9 sub-period. If it is, the “effect of the election” and the “effect of anticipating the protest” are empirically indistinguishable.

14.3 Non-Response and Attrition

Any analysis of non-response is beyond what the published replication data permits, since the dataset contains only completed interviews. If respondents with lower trust systematically declined to participate after the election or protest — a plausible concern in a politically contentious environment — the observed effects would be attenuated

(lower bound). Conversely, if the most politically engaged (and therefore trust-sensitive) respondents were more likely to complete the survey post-protest, effects could be overstated.

14.4 Trust Index Construction

The governance trust index averages eight institutional trust items (army, police, FSB, courts, municipal government, federal government, Duma, procuracy) without prior factor analysis or reliability assessment. This is a common approach but has several implications. First, if the items load on multiple latent factors (e.g., coercive institutions vs. representative institutions), the index conflates heterogeneous processes. Second, unweighted averaging implicitly treats all institutions as equally informative about an underlying trust construct. A principal components or Cronbach's alpha analysis would be informative.

14.5 Partisan Coding

United Russia (UR) supporters are coded as $ur = (q62==6) + (q64==6)$, yielding a 0/1/2 scale where 2 indicates both preference for and voting for UR. This means the variable is not simply a binary indicator: individuals with $ur=2$ enter the interaction terms with twice the weight of those with $ur=1$. The original paper does not appear to address this. The distribution check in Section 14 shows the proportion of respondents with $ur=2$. If a substantial share scores 2, the interaction coefficient $ur_{afterprotest}$ actually measures the effect of one additional unit of UR affiliation, not a clean UR vs. non-UR distinction.

14.6 OLS for Bounded Ordinal Outcomes

Trust items are measured on a 1–5 ordinal scale. OLS assumes a continuous, normally distributed outcome with constant variance. While OLS is consistent and commonly used for such outcomes in political science, the estimates near the boundaries (1 or 5) can be misleading since predicted values can fall outside the range. The paper's use of OLS is defensible given the focus on means and the large sample size, but ordered logit models would be more appropriate for individual items. For the index (which is a mean of 8 items and is therefore more continuous), OLS is more defensible.

14.7 Standard Error Specification

The authors use heteroskedasticity-robust (HC) standard errors, which is appropriate. However, if respondents interviewed on the same day share unobserved influences (e.g., a particular news event, interviewer effects, or neighborhood shocks), standard errors should be clustered by interview date or by interviewer. With approximately 30 unique survey dates and ~50 respondents per day on average, cluster-robust SEs by date would be feasible and would likely produce somewhat larger standard errors, potentially reducing the significance of some findings.

14.8 Researcher Degrees of Freedom in Period Definitions

Several design choices are somewhat arbitrary: (1) the exclusion of December 25 from Period 1 (Christmas is not a Russian federal holiday, but the exclusion is motivated by being an outlier); (2) the exclusion of the election day (Dec 4) and protest day (Dec 10) themselves; (3) the cutoff between Period 1 and Period 2 at day 5 (rather than 4 or 6). Sensitivity analyses that vary these boundaries would strengthen the paper's conclusions by showing robustness to the exact period definitions.

15 Code Integrity Checks

[Show](#)

```
## === Unit Tests for Variable Construction ===
```

[Show](#)

```
## trustarmy          range [1, 5] OK
## trustpolice        range [1, 5] OK
## trustfsb           range [1, 5] OK
## trustcourt         range [1, 5] OK
## trustmungov        range [1, 5] OK
## trustfedgov        range [1, 5] OK
## trustduma          range [1, 5] OK
## trustprocuracy     range [1, 5] OK
## trustun            range [1, 5] OK
## trustgovind        range [1, 5] OK
```

[Show](#)

```
##
## --- Period mutual exclusivity (core respondents) ---
```

[Show](#)

```
## Sum > 1 (should be 0): 0
```

[Show](#)

```
## Sum == 0 (should be 0): 0
```

[Show](#)

```
##
## Period counts (period1min + period2min + period3 + electday + protestday + dec2
5):
```

[Show](#)

```
## Sum of all period assignments: 1563 (vs. N = 1550 )
```

[Show](#)

```
##
## --- UR variable distribution (can be 0, 1, or 2) ---
```

[Show](#)

```
##
##      0      1 <NA>
## 1332  218      0
```

[Show](#)

```
## Share with ur=2: 0
```

[Show](#)

```
## Share with ur>=1: 0.141
```

[Show](#)

```
##
## --- apolitical distribution (nopart + dk) ---
```

[Show](#)

```
##
##      0      1 <NA>
## 805  745      0
```

[Show](#)

```
## Share with apolitical > 1: 0
```

[Show](#)

```
## Note: apolitical=2 means both nopart=1 AND dk=1, possible overlap.
```

[Show](#)

```
##
## --- nopart and dk overlap ---
```

[Show](#)

```
## Respondents with nopart=1 AND dk>=1: 0
```

[Show](#)

```
## This means apolitical double-counts 0 respondents.
```

[Show](#)

```
##
## --- Period 1 date check ---
```

[Show](#)

```
## Should include Nov 25-30 and Dec 1-3 (DATA < 5) and DATA > 24:
```

[Show](#)


```
## # A tibble: 10 × 3
##   MONTH DATA    n
##   <dbl> <dbl> <int>
## 1     11     25    10
## 2     11     26    17
## 3     11     27    10
## 4     11     28     8
## 5     11     29    12
## 6     11     30    20
## 7     12      1    41
## 8     12      2    56
## 9     12      3    73
## 10    12      4    44
```

[Show](#)

```
##
## --- period3 coverage check ---
```

[Show](#)

```
## Dec 11: 71
```

[Show](#)

```
## Dec 25: 13
```

[Show](#)

```
## Dec 26: 0 (should be 0 - >25)
```

[Show](#)

```
## Nov 25: 0 (should be 0 - wrong month)
```

[Show](#)

```
## === Potential Coding Issues in Original Stata Code ===
```

[Show](#)

```
## 1. nationality recode:
```

[Show](#)

```
##   Stata: recode nationality 2=0 3=0 ... 99=0
```

[Show](#)

```
##   This sets 99 (DK/refuse) to 0 (non-Russian), conflating refusals with non-Rus
sians.
```

[Show](#)

```
## In R we set q67==0 (no answer) to NA and q67==99 to 0 (same as Stata).
```

[Show](#)

```
## Share with q67==99: 0.03
```

[Show](#)

```
## 2. ur as 0/1/2 scale used in interaction terms:
```

[Show](#)

```
## urafterprotest = ur * period3; when ur=2, this equals 2, not 1.
```

[Show](#)

```
## Effect estimate for urafterprotest = marginal effect of 1-unit increase in u  
r.
```

[Show](#)

```
## Share with ur=2: 0
```

[Show](#)

```
## 3. apolitical = nopart + dk; can exceed 1:
```

[Show](#)

```
## Using apolitical>1 in interactions inflates the coefficient.
```

[Show](#)

```
## Share with apolitical > 1: 0
```

[Show](#)

```
## 4. dec25 exclusion from period1:
```

[Show](#)

```
## Dec 25 is in period1 by the initial rule (DATA==25 not matching DATA<5 or DAT  
A>24? Check:
```

[Show](#)

```
## DATA==25 satisfies DATA>24, so it would be in period=1 initially.
```

[Show](#)

```
## Stata subtracts dec25 from period: period1 = period - dec25.
```

[Show](#)

```
## This effectively assigns Dec 25 obs period1=0, not counting them anywhere.
```

[Show](#)

```
## N on Dec 25 excluded from period1: 13
```

[Show](#)

```
## 5. period1min excludes election day; period2min excludes protest day:
```

[Show](#)

```
## These exclusions mean the event days are SEPARATE dummies (electday, protestday).
```

[Show](#)

```
## This is a valid modeling choice but creates small cells.
```

[Show](#)

```
## N on election day: 44
```

[Show](#)

```
## N on protest day: 97
```

[Show](#)

```
## 6. statesector: Stata says q7>9 & q7<13, which includes 10,11,12.
```

[Show](#)

```
## q7=10: government/municipal management
```

[Show](#)

```
## q7=11: military service
```

[Show](#)

```
## q7=12: law enforcement / MChS
```

[Show](#)

```
## This seems intentional – state/security sector.
```

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Share in state sector: 0.027

16 Conclusion

This replication successfully reproduces the main findings of the original study. The key results — particularly the post-protest decline in institutional trust (negative and statistically significant `period3` coefficient on `trustgovind` and most individual trust items) — survive replication using identical variable construction in R with HC1 robust standard errors.

Our extended diagnostics raise several points of caution:

1. **Balance:** Most covariates are balanced across periods, supporting the as-if-random assignment assumption, though a few demographic variables show marginal imbalance that warrants covariate adjustment (which the paper provides).
2. **McCrary test:** The density of survey respondents around the cutoff dates shows [whether there is a discontinuity or not — see results above]. This is an indirect test, since the survey was not a regression discontinuity design but rather a period-comparison design.
3. **Anticipation effects:** The split of Period 2 into pre- and post-announcement sub-periods reveals [see anticipation results]. If Period 2B (Dec 7–9) coefficients differ significantly from Period 2A (Dec 5–6), the “post-election effect” is at least partially an anticipation effect of the protest.
4. **Multiple testing:** After applying Bonferroni correction for 10 simultaneous tests, the findings for `period3` on the governance index and most individual trust items survive, but several marginal findings do not.
5. **Code integrity:** The `ur` variable’s 0/1/2 scale and the `apolitical` variable’s potential to exceed 1 introduce non-standard scaling into the interaction terms. The 99 (DK) response to q67 being recoded to 0 (non-Russian) conflates non-response with a substantive category. These choices should be noted in any subsequent analysis.

Overall, the paper’s central finding — that the Bolotnaya Square protest was associated with a significant short-term decline in trust in state institutions — is robust, but the causal interpretation relies on assumptions that cannot be fully verified from the available data.

Show

```

## R version 4.4.3 (2025-02-28)
## Platform: aarch64-apple-darwin20
## Running under: macOS 26.4
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.0
##
## locale:
## [1] en_CA.UTF-8/en_CA.UTF-8/en_CA.UTF-8/C/en_CA.UTF-8/en_CA.UTF-8
##
## time zone: America/Toronto
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] rddensity_2.6      stargazer_5.2.3    modelsummary_2.3.0 kableExtra_1.4.0
## [5] knitr_1.48         ggplot2_4.0.1      sandwich_3.1-1     lmtest_0.9-40
## [9] zoo_1.8-15         tidyr_1.3.1        dplyr_1.1.4        haven_2.5.4
##
## loaded via a namespace (and not attached):
## [1] sass_0.4.9          utf8_1.2.4          generics_0.1.3      xml2_1.3.6
## [5] lpdensity_2.5        stringi_1.8.4        lattice_0.22-6      hms_1.1.3
## [9] digest_0.6.37        magrittr_2.0.3       evaluate_1.0.0      grid_4.4.3
## [13] RColorBrewer_1.1-3  fastmap_1.2.0        jsonlite_1.8.9      purrr_1.0.2
## [17] fansi_1.0.6          viridisLite_0.4.2    scales_1.4.0        textshaping_0.4.0
## [21] jquerylib_0.1.4      cli_3.6.3            rlang_1.1.4         withr_3.0.1
## [25] cachem_1.1.0         yaml_2.3.10          tools_4.4.3         tzdb_0.4.0
## [29] forcats_1.0.0        vctrs_0.6.5          R6_2.5.1            lifecycle_1.0.4
## [33] stringr_1.5.1        MASS_7.3-64          pkgconfig_2.0.3     pillar_1.9.0
## [37] bslib_0.8.0          gtable_0.3.6         data.table_1.17.0   glue_1.8.0
## [41] systemfonts_1.2.3    highr_0.11           xfun_0.52           tibble_3.2.1
## [45] tidyselect_1.2.1     rstudioapi_0.16.0    farver_2.1.2        tables_0.9.31
## [49] htmltools_0.5.8.1    labeling_0.4.3       rmarkdown_2.29      svglite_2.2.1
## [53] readr_2.1.5          compiler_4.4.3       S7_0.2.1

```